

Math 133: Quiz 3 — Draft Key

September 25, 2026

1. (a) $\frac{C(15) - C(10)}{15 - 10} = \frac{135 - 100}{5} = 7$ dollars per item.

(b) $C(11) - C(10) = 106 - 100 = \6 .

2. The model is $A(t) = 60(1.04)^t$, with t in years. Solving $60(1.04)^t = 100$ gives

$$t = \frac{\ln(100/60)}{\ln(1.04)} \approx 13.0 \text{ years.}$$

3. (a) $f(-2) = 4$ and $f(0) = 0$, so $m = \frac{0 - 4}{0 - (-2)} = -2$. The secant line is $y = -2x$.

(b) Since $f(0) = 0$ and $f(4) = 1$, the net change is $f(4) - f(0) = 1$. The average rate of change is $\frac{f(4) - f(0)}{4 - 0} = \frac{1}{4}$.

(c)

$$\lim_{x \rightarrow -1} f(x) = 1, \quad \lim_{x \rightarrow 1^+} f(x) = 4, \quad \lim_{x \rightarrow 1^-} f(x) = 1, \quad \lim_{x \rightarrow 1} f(x) \text{ does not exist.}$$

The two-sided limit at 1 does not exist because the one-sided limits differ.